

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			Diagram to show insulated / polystyrene container with reagents (1) Lid (1) Thermometer (1)		3		3		3
	(b)			Energy = $25 \times 4.18 \times 14.5 = 1515 \text{ J}$ (1) Moles Mg = $0.1/24.3 = 4.12 \times 10^{-3}$ (1) $\Delta H = 1515/ 4.12 \times 10^{-3} = -368 \text{ kJmol}^{-1}$ (1) ecf possible		3		3	3	
	(c)			Moles Mg = 8.23×10^{-3} (1) Requires 1.64×10^{-2} moles HCl for all to react Moles HCl = 5×10^{-3} i.e. not enough for a complete reaction (1)			2	2		
	(d)			Maximum error in the one reading taken = 0.05 g (1) (Allow 0.1 g if clearly states that mass was measured using weighing by difference) Maximum % error = $\frac{0.05}{0.1} \times 100 = 50\%$ (1)		1		2	1	1
	(e)			Cannot measure ΔH – can only measure ΔT / no water / solution present to measure temperature change			1	1		1
Question 7 total					0	7	4	11	4	5